

FRIENDLY BEACHES, Australia

WMO: 949870

Lat: 41.995S

Lon: 148.279E

Elev: 184

StdP: 14.60

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.2	38.9	28.4	22.5	47.5	30.1	24.3	46.2	24.3	52.2	21.8	52.7	7.5	290	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.1	80.3	64.0	76.5	63.9	73.8	62.7	67.7	75.6	66.2	73.0	64.8	71.0	12.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.0	70.2	63.8	89.1	68.8	62.2	84.4	67.8	32.3	75.7	31.0	73.1	29.9	71.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	20.8	18.5	33.3	93.1	1.4	5.6	32.3	97.1	31.5	100.4	30.7	103.6	29.6	107.7
			32.4	71.2	1.1	1.6	31.6	72.4	31.0	73.3	30.4	74.2	29.6	75.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	63.9	63.6	61.7	57.3	53.5	50.2	49.5	50.1	52.9	55.2	58.0	61.2	(6)
	DBStd	6.70	4.68	4.55	4.68	4.60	4.32	3.69	3.60	3.97	4.20	4.75	4.59	4.56	(7)
	HDD50	184	0	0	0	2	15	43	54	48	16	5	1	0	(8)
	HDD65	3273	76	73	124	235	357	445	481	461	363	307	215	136	(9)
	CDD50	2516	430	380	363	222	122	48	38	52	103	168	242	348	(10)
	CDD65	129	41	33	22	4	0	0	0	0	0	4	6	19	(11)
	CDH74	692	231	150	88	16	0	0	0	0	2	41	58	106	(12)
	CDH80	178	68	37	17	2	0	0	0	0	0	9	17	28	(13)
(14) Wind	WSAvg	9.0	10.2	9.3	9.3	8.1	8.2	7.9	8.3	8.9	9.2	9.5	9.8	10.0	(14)
(15) Precipitation	PrecAvg	28.9	2.4	1.8	2.2	2.6	1.9	1.9	2.3	2.3	2.0	2.3	2.5	3.3	(15)
	PrecMax	49.6	9.5	7.6	9.4	14.8	13.2	7.6	7.3	8.2	6.5	8.1	8.2	13.3	(16)
	PrecMin	14.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3	0.0	0.0	0.1	0.0	(17)
	PrecStd	9.0	2.4	1.7	1.9	2.6	2.4	1.6	1.9	1.8	1.7	1.7	1.9	3.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.7	85.3	81.4	75.8	67.8	62.2	61.7	65.4	70.3	79.2	81.3	84.1	(19)
		MCWB	65.4	66.5	64.7	60.8	57.4	54.1	52.3	53.0	56.2	57.8	60.3	63.7	(20)
	2%	DB	79.2	77.9	75.8	71.0	65.1	60.3	59.4	61.9	66.1	71.0	73.0	76.1	(21)
		MCWB	65.1	65.9	64.7	60.5	56.1	53.3	51.2	52.2	54.2	57.2	60.2	62.6	(22)
	5%	DB	75.2	74.5	72.6	67.9	63.0	58.7	57.6	59.3	63.2	66.9	69.3	72.7	(23)
		MCWB	64.1	64.4	63.0	59.0	55.0	52.3	50.2	51.0	53.2	56.0	58.8	61.4	(24)
	10%	DB	72.5	71.8	69.9	65.4	61.1	56.9	56.0	57.3	60.8	63.8	66.7	69.9	(25)
		MCWB	62.3	62.5	61.5	57.9	54.0	50.9	49.5	49.9	52.0	54.4	57.5	60.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	69.5	69.0	64.8	60.3	58.2	54.8	55.6	58.7	61.8	64.7	67.3	(27)
		MCDB	78.5	77.4	76.8	70.2	62.9	59.3	58.0	59.9	65.6	69.4	73.8	78.1	(28)
	2%	WB	67.4	67.5	66.6	62.5	58.2	54.9	52.9	53.8	56.4	59.3	62.3	64.9	(29)
		MCDB	75.7	75.0	72.4	67.6	62.5	58.8	56.9	58.9	62.6	66.4	70.1	72.7	(30)
	5%	WB	65.6	66.0	64.9	60.8	56.4	52.9	51.6	52.3	54.8	57.6	60.6	63.1	(31)
		MCDB	73.2	72.0	70.1	66.0	61.6	57.4	56.1	57.8	60.8	64.4	67.4	70.0	(32)
	10%	WB	63.9	64.5	63.1	59.1	54.8	51.4	50.3	50.9	53.3	55.9	58.9	61.4	(33)
		MCDB	70.8	69.9	68.4	64.3	60.3	56.1	55.2	56.5	59.3	62.2	64.9	67.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	15.3	14.8	14.2	12.5	12.0	12.3	13.3	15.2	15.8	15.2	15.8	(35)
		MCDBR	20.8	19.2	17.7	17.4	14.1	13.3	13.3	15.1	17.4	20.1	19.7	20.0	(36)
		MCWBR	11.1	10.7	10.3	10.7	9.5	9.1	8.7	9.6	10.6	11.1	10.7	10.9	(37)
	5% WB	MCDBR	19.1	16.4	15.5	15.2	13.0	11.9	12.1	14.0	15.6	17.6	17.5	17.2	(38)
		MCWBR	10.9	9.9	9.9	10.3	9.6	8.7	9.0	9.6	10.4	10.9	10.4	10.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.346	0.342	0.327	0.317	0.301	0.297	0.290	0.307	0.324	0.333	0.340	0.340	(40)
	taud		2.536	2.564	2.606	2.603	2.613	2.605	2.636	2.588	2.535	2.523	2.532	2.542	(41)
	Ebn at Noon		311	304	295	276	259	248	261	275	291	303	311	315	(42)
	Edh at Noon		34	31	28	24	20	18	19	23	29	32	34	34	(43)
(44) All-Sky Solar Radiation	RadAvg		2096	1763	1388	948	621	491	564	824	1236	1637	1889	2147	(44)
	RadStd		146	83	65	61	36	39	27	55	82	84	140	120	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	+0.56	N/A	N/A	+1.35	N/A	N/A	N/A	-185	+198	+32

Nomenclature: See separate page